POWERING POTENTIAL

Software Developer

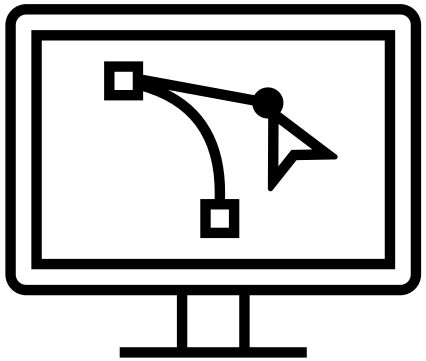
Agile Software Development

The SDLC

What is systems development?

The process of taking a set of business requirements through a series of structured stages and translating them into an operational IT system.

Let’s see what these structured stages are …



What is a systems development life cycle?

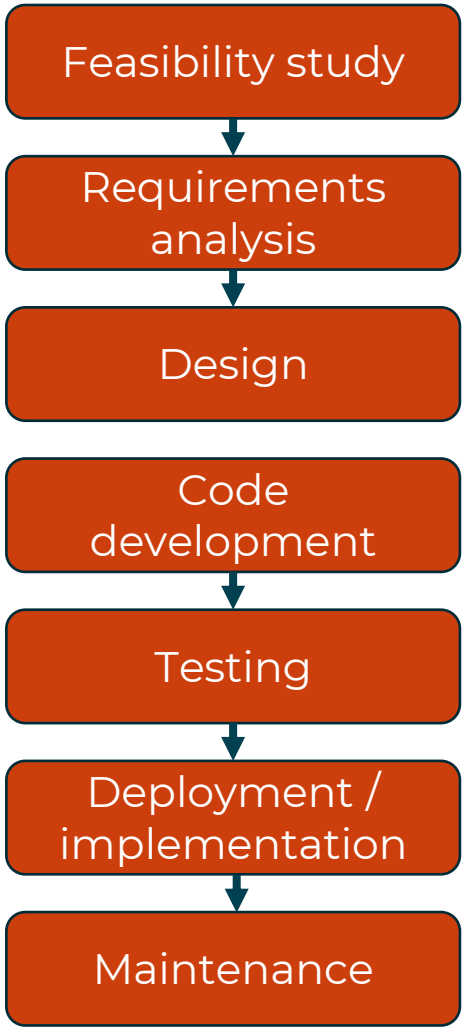
A system development life cycle (SDLC) is a framework describing a process to:

- understand
- plan
- build
- test
- deploy

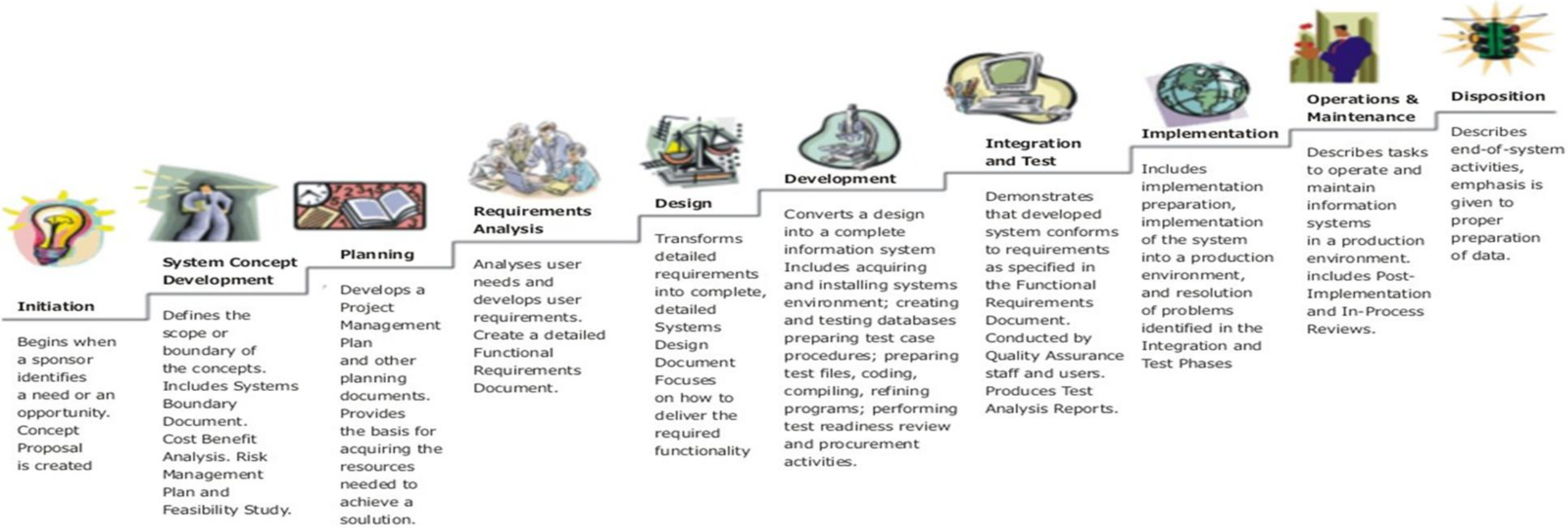
an information system.

Can apply to:

- Hardware
- Software
- Both



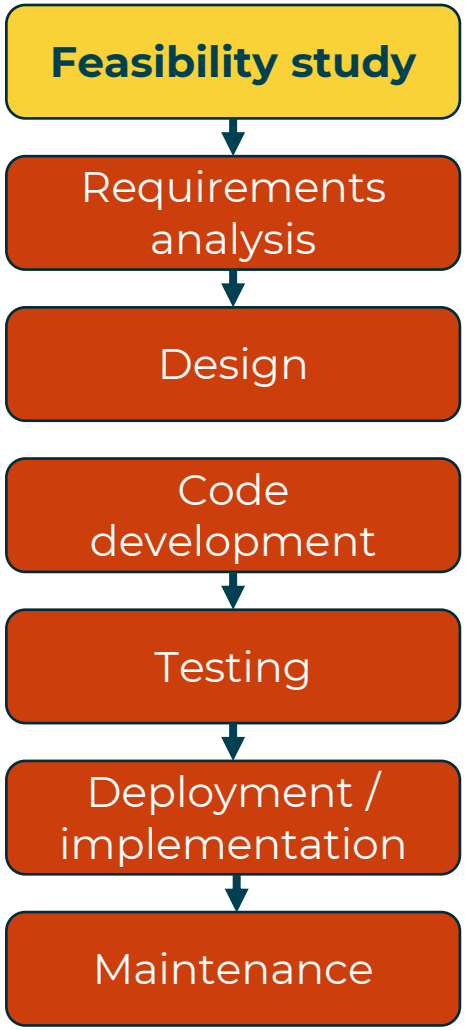
Life cycle phases



Feasibility (initiation) **study**

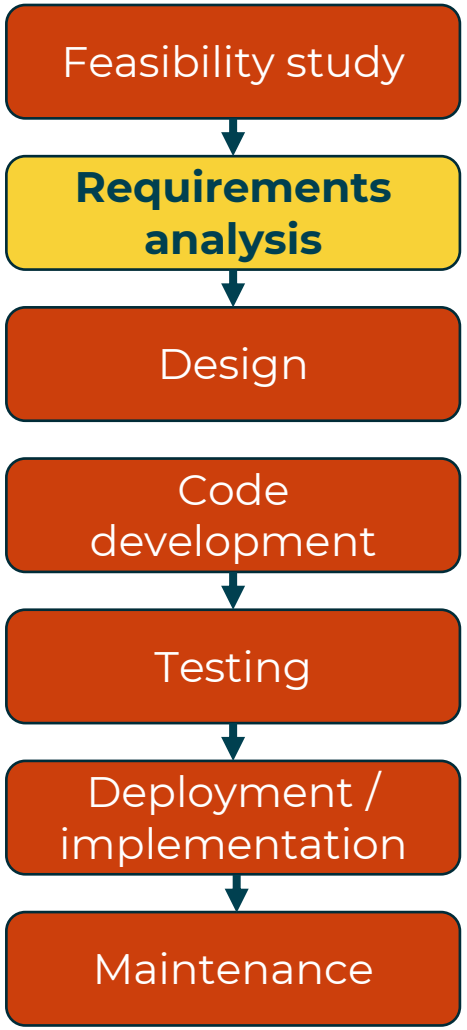
- **Funding** required before starting.
- Evaluate benefits and likelihood of success.
- Resources and costs versus value of completed system:
  - Business case
  - Return on investment (**ROI**)

- Time dependent on size and complexity of project.



Requirement analysis

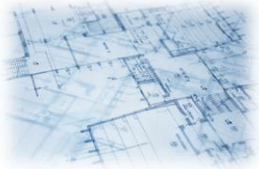
- What business needs proposed system to do.
  - Ask if unsure.
- Elicit, analyse, document, and validate requirements.
- Decide how to store, manage, access, and update requirements.
- Key input to design; requirements traced through SDLC.
- Level of detail influenced by SDLC.
- Also known as **requirements engineering**.



The importance of good requirements

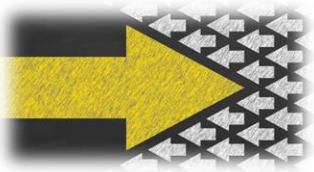
What do you think is the consequence of badly/poorly designed system requirements?

Characteristics of good requirements

Categorised	Grouped by type of requirement to make easier to review
Relevant	Within scope of the project. 
Prioritised	
Achievable	Within the technical or budget constraints
Unambiguous	Unambiguous and clearly expressed
Testable	
Requirement not solution	

Characteristics of good requirements

Consistent	No conflict with other requirements
Owned	By someone who can make decisions
Unique and atomic	About one thing
Traceable	Where requirement came, where to go for more info
Conformant	Meets standards of how requirements should be documented



Requirement Classification

Functional

Define **what** the solution must do

For example to identify an invoice is overdue and issue alert

Non-Functional

Define **how** the solution must operate (product qualities)

For example the result of a query must be displayed with two seconds of pressing enter

General

High-level overarching requirements of a business policy nature

For example to comply with data protection

Technical

High-level overarching requirements of a technical policy nature

For example to operate on a certain platform and use a particular programming language

Functional Requirements - categories

Is **What** the system is expected to do

- Data entry – Gathering and recording data
- Data maintenance
  Changes to data, including data deletion
- Procedural – Implementation of business rules
- Retrieval – Reporting, responding to enquiries

Non-functional requirements

Define constraints on **how** the functional requirements will be provided

→ Factors in the cost and are very important to the success of the product

→ Define a characteristic we must implement

Non-functional requirements

Availability - System is available when required to fulfil its purpose

- E.g. Reprint the report on demand
  This is an availability constraint – When this functionality must be available

Capacity - Supports transaction and other data volumes as necessary

Throughput – Relates to data/transactional volumes within a specific timeframe

- E.g. Respond to a credit enquiry within 5 seconds. This is a performance constraint

Scalability - Scale up/down the number of transactions or users

Reliable – System is robust and available as defined by SLA.

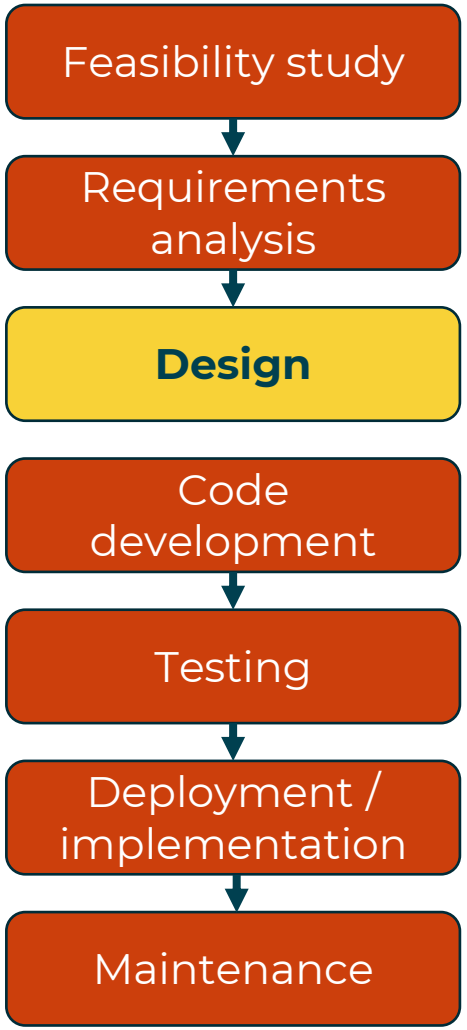
Maintainability - How easy to fix/change – Designed well, flexible, well-documented

Non-functional requirements types

- **User-friendly** Functionality delivered in a consistent, intuitive way.
- **Secure** Adequate controls to restrict access to the system and its data.
- **Accessible** Is accessible to all users regardless of disability, language...
- **Responsive** Provides a response to a particular request within a timeframe.
- **Flexible** Adapt to changing business situations.
- **Portable** Can ‘port’ the design to different technology platforms/devices.
- **Configurable** Can change the behaviour by setting configuration options

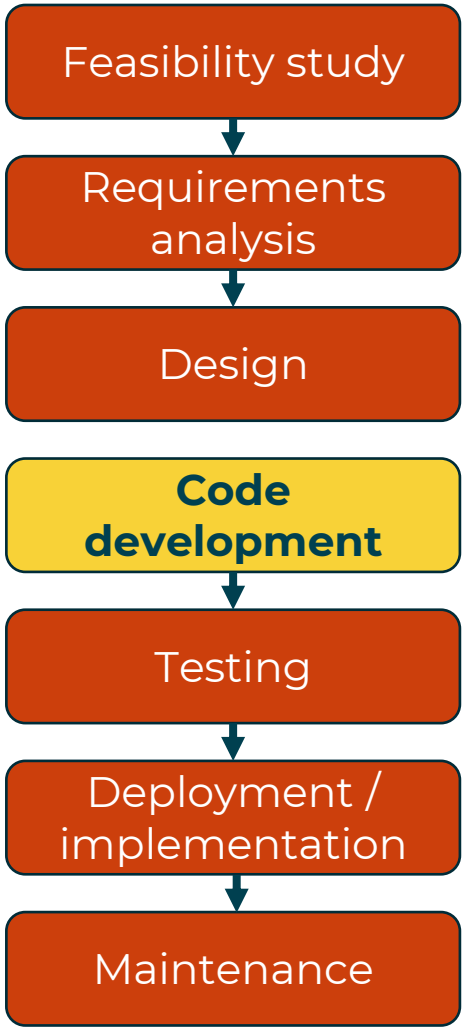
Design

- Evaluate possible solutions that meet requirements.
- Chosen design elaborated to sufficient detail to start development.



Code development

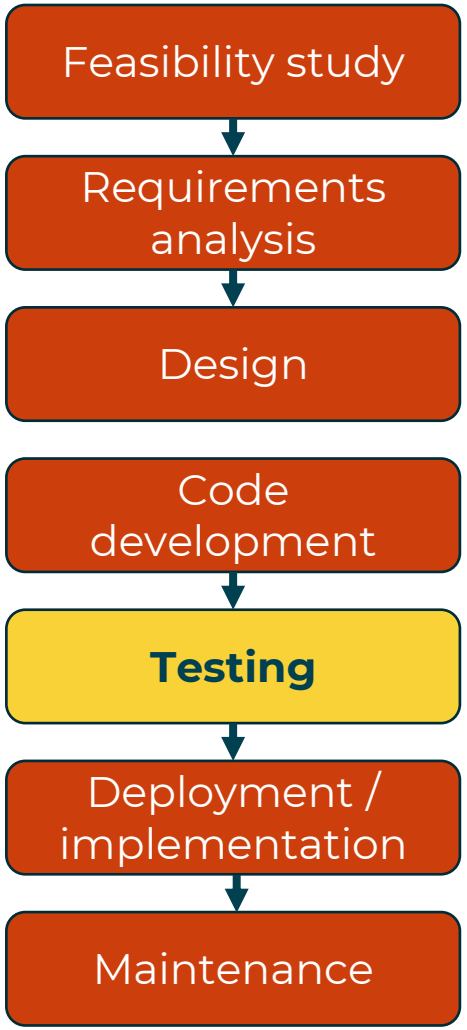
- Hardware and software technical components created, procured or configured
- Follow design to ensure system does what is required
- Also known as **programming** or **build**.



Testing

- Components produced tested to ensure working properly and does what supposed to do.

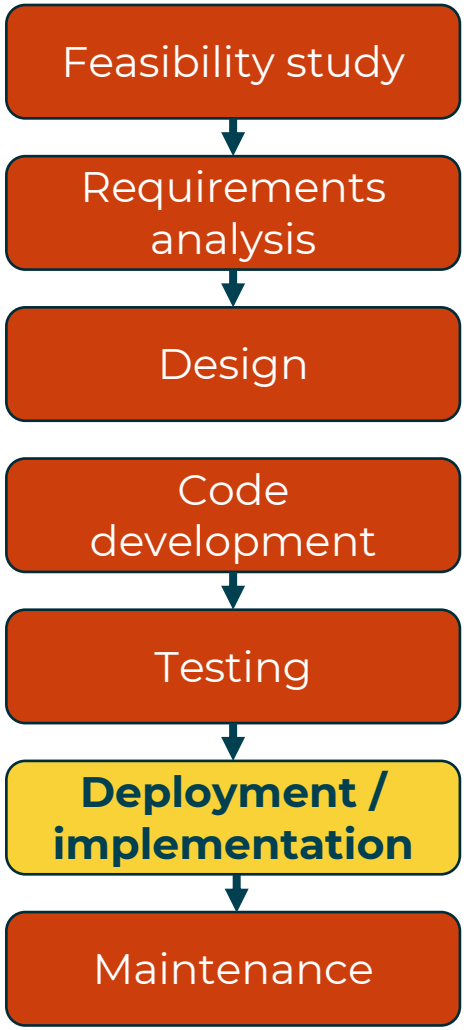
- **Different levels of testing:**
  - Unit
  - Integration
  - System
  - User acceptance



Deployment / implementation

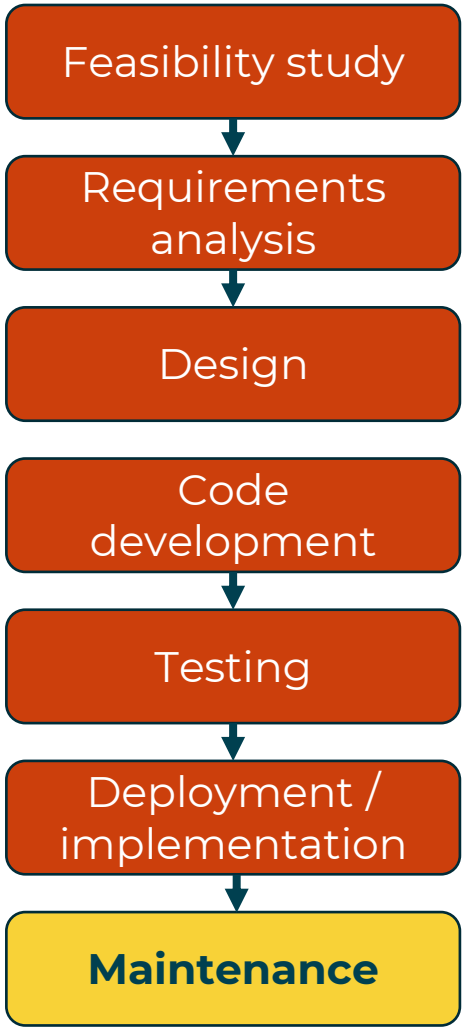
- Commissioning in a live environment:
  - Also known as **operation** or **production** environment.
  - Developed in test environment to protect live systems.

- Needs to be carefully planned, understood, and managed.



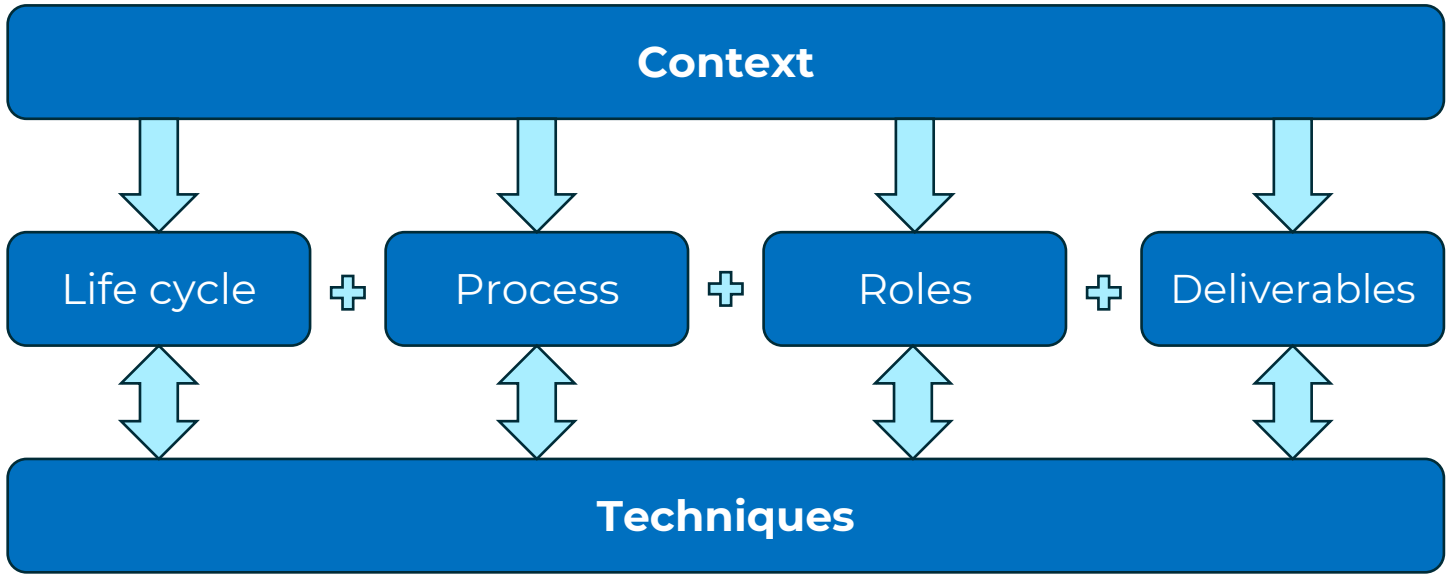
Maintenance

- More than just correcting faults or keeping the system in good running order:
  - 20% corrective
  - 80% enhancements
- Depends on:
  - development strategy and lifetime of operation
- **Maintenance types:**
  - Corrective
  - Adaptive
  - Perfective
  - Preventative

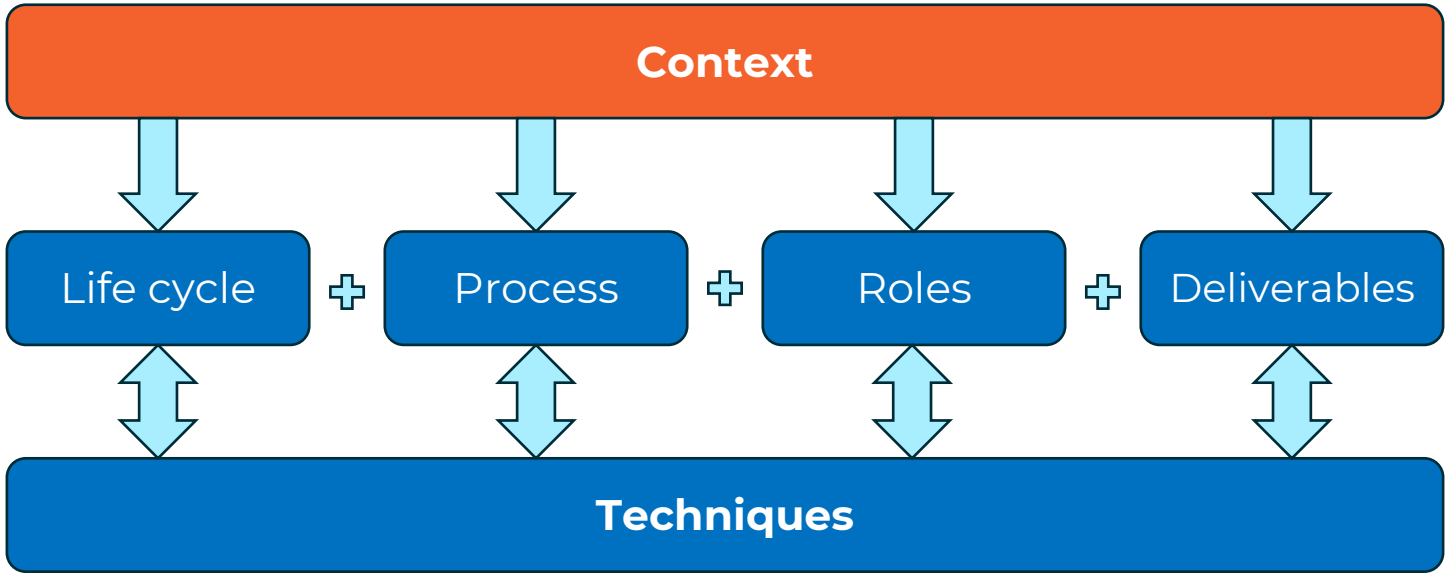


Elements of the SDLC

In addition to SDLC main stages, you also need to consider other elements:



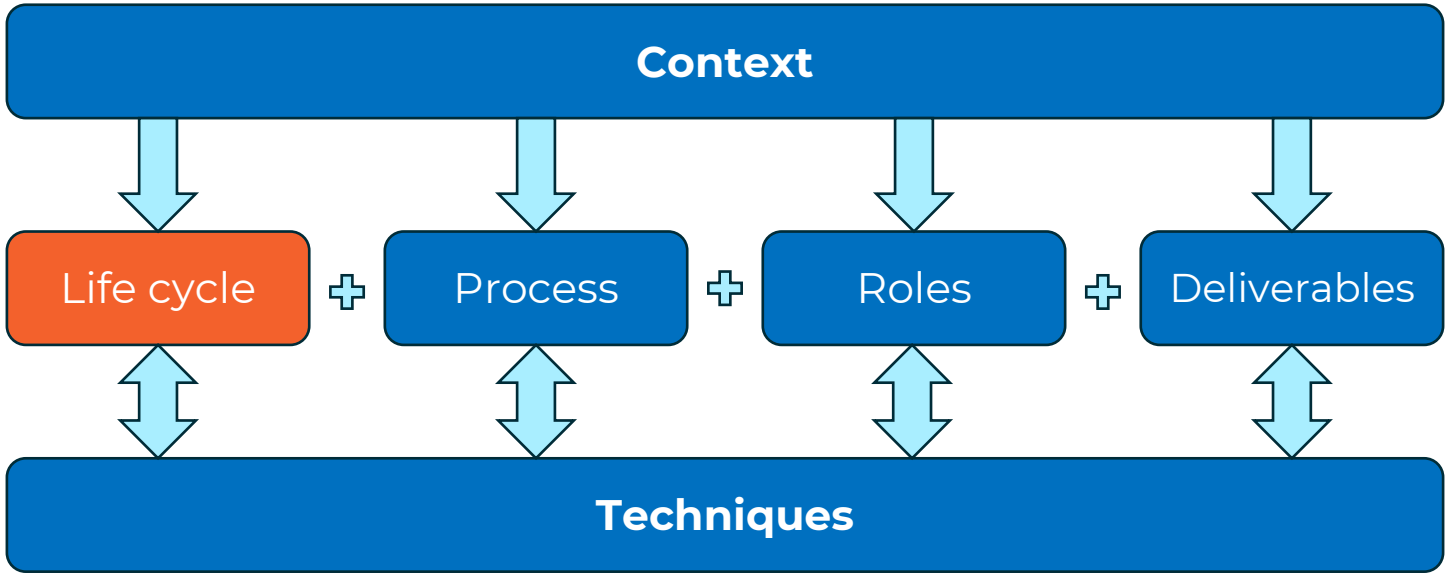
Context



Needs to be considered before starting:

- **Release plan**: single or multiple
- Level of **skills** and expertise: preferred delivery style
- **Location of teams** and stakeholders: single site or dispersed
- **Requirements**: audit, quality, regulatory, complexity and stability
- **Technology** to be used: tried and tested or new

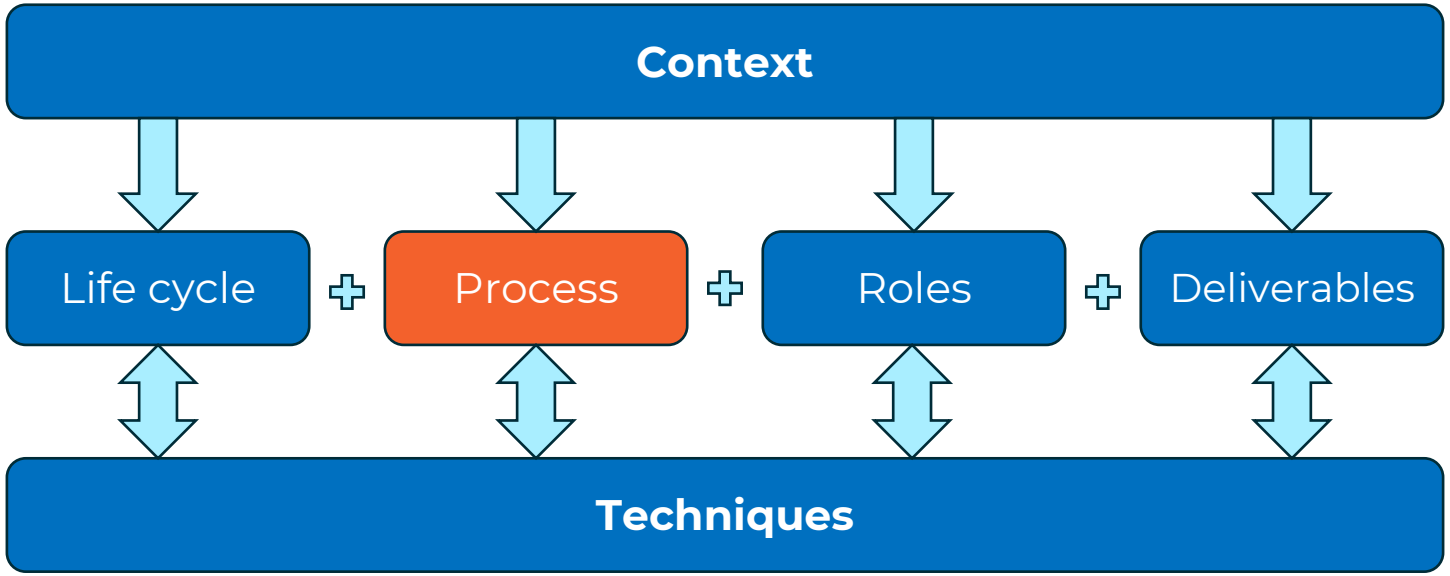
Life cycle



- Describes stages to follow: plan, design, build, test, and deliver.

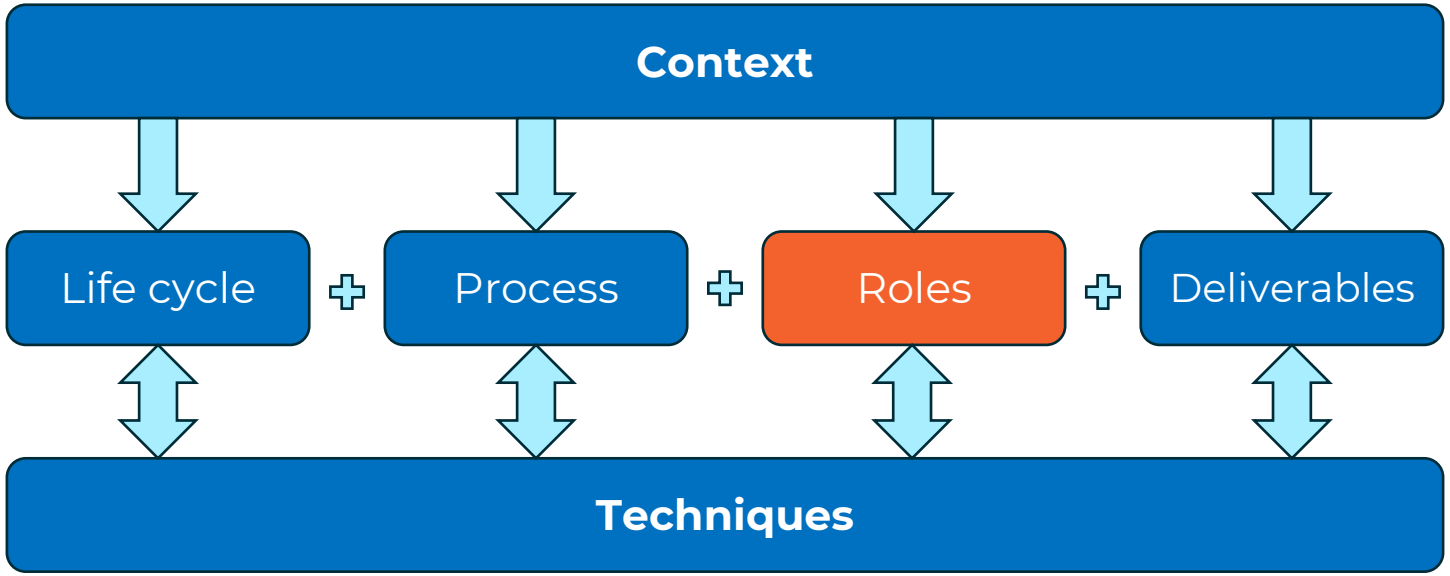
  - **Linear** sequential or step by step
  - **Evolutionary** iterative, evolving through progressive versions

Process



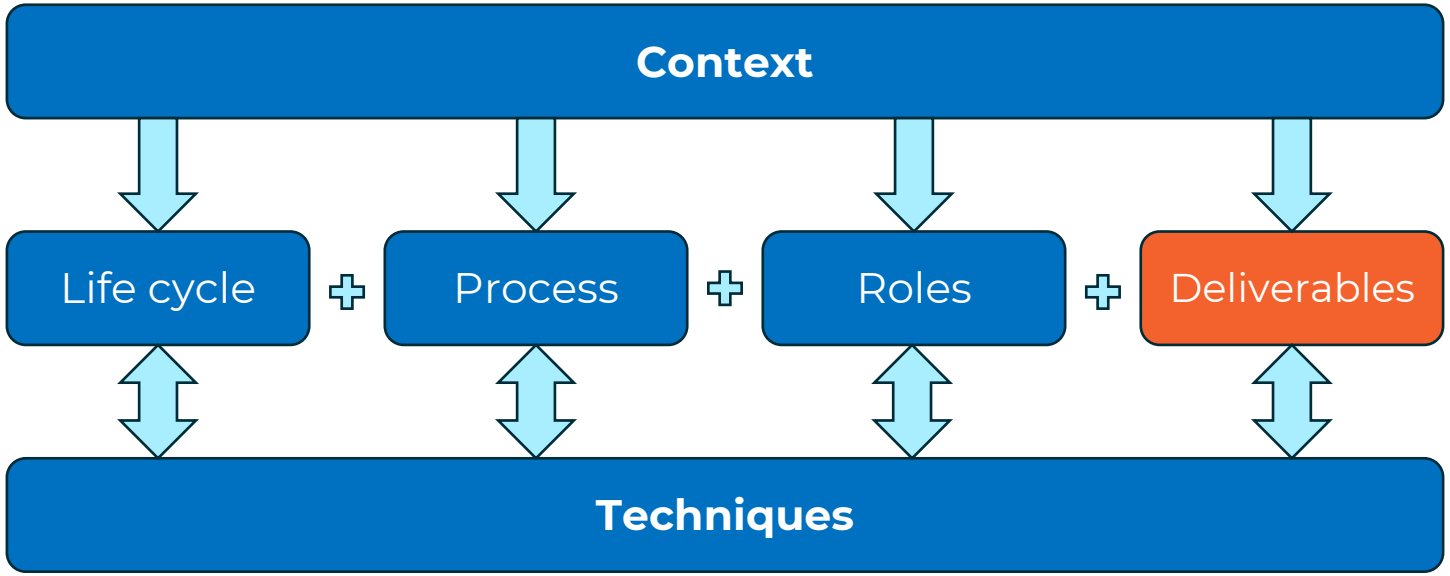
- Set of steps and actions followed to deliver outcomes within stages.
- Methods and standards support the framework within SDLC:
  - Include defined roles, deliverables, techniques, models
  - Prescriptive: specific SDLCs
  - Agnostic: various SDLCs
  - Dependent on context and lifecycle chosen

Roles



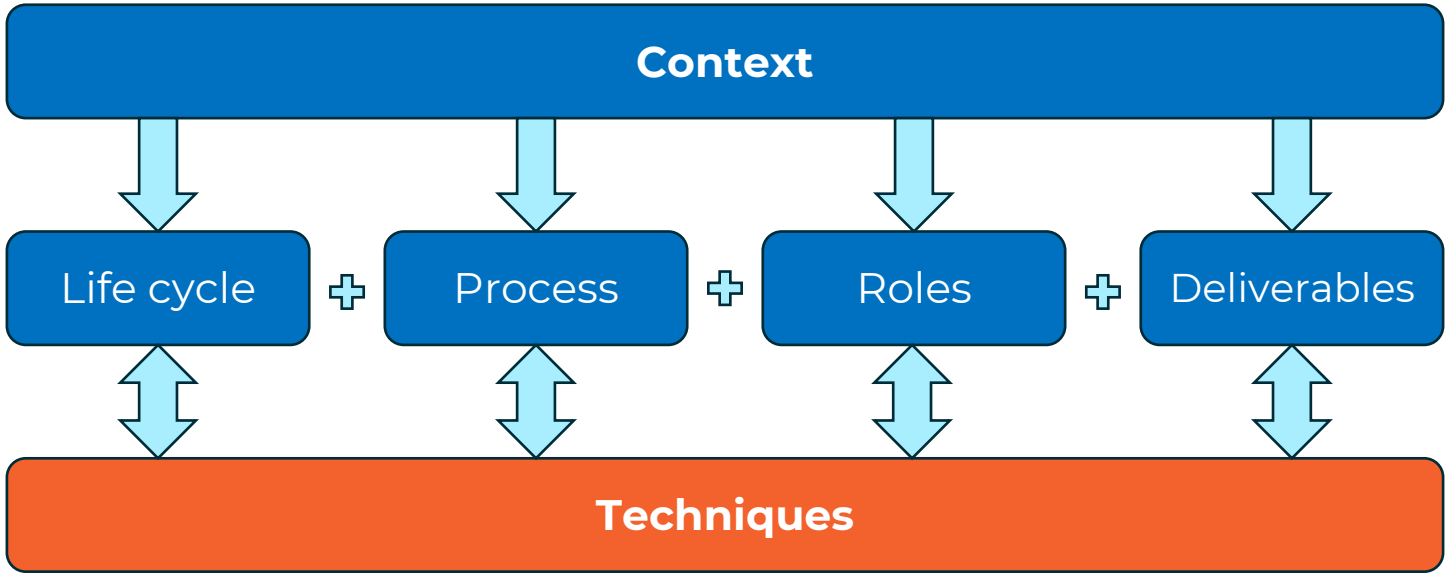
- People that carry out tasks within various SDLCs:
  - Specific to particular SDLC or generic: various titles
  - Include business, project, technical, and support roles

Deliverables



- **Documents, models, designs, hardware and software required**:
  - Vary for different SDLCs and stages
- Detailed understanding of requirements – what, how, level of quality, when?
- Examples:
  - Requirement documents, models, system / software components
  - Test / implementation plans and scripts

Techniques



- **Numerous**: choice dependent on team, organisation, SDLC. For example:
  - Military: Ministry of Defence Architecture Framework (MoDAF)
  - Commercial payment milestones and reviews: **Waterfall**
  - Rapid prototypes: **Agile**

- Many SDLCs: strengths / weaknesses, suit particular projects

WHAT ARE DELIVERABLES?

Outputs / documents from each stage of the SDLC:

- Review, sign-off, and become inputs for following stage.
- Type, format, and content dependent on methodology adopted.
- Linear vs. evolutionary.
- Also known as **system documentation**.

DELIVERABLES

Feasibility study (initiation):

- Project brief / charter / initiation / scope
- Terms of reference
- Business case: cost / benefit analysis
- System scope / requirements
- High level plan (Gantt chart)

Requirements analysis:

- Requirements report
- Threats and risks report
- Recommendations

Design:

- Detailed technical specifications
- Logical and physical designs

DELIVERABLES

Code development:

- Developed system: software and hardware if appropriate
- Code / screens / reports
- Database
- Interfaces

Testing:

- Tested system
- Test plans / cases / specifications

DELIVERABLES

Deployment / implementation

- Implemented system
- User documentation
- Training procedures
- Support capabilities

Maintenance:

- How to guides
- Completed tickets

General considerations

Documentation meets organisational standards:

- Documentation content and layout
- Naming conventions
- Diagrams
- Quality control requirements

Organisational processes and requirements:

- Project management
- Governance

review

Quiz

POWERING POTENTIAL

QUESTIONS AND FEEDBACK